LESSON PLAN · SCRATCH (GR 3–5) ADVANCED: ARCADE ENGINEERS

Lesson 11: Game States and Levels

Module 4 | 45 minutes | Game states | Grades 3–5 · ages 9–11

Objective: Use broadcasts and variables to add a start screen, levels and a game-over state to a game.

Success criteria

- I can add a start menu with a broadcast.
- I can change levels with a variable.
- I can plan the states of my game.

Key vocabulary

- **state**: What part of the game you are in right now.
- **level**: A stage of a game, usually harder each time.
- **game over**: The state when the game ends.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Traffic-light game (5 min): a learner shows red/amber/green cards; the class acts out what each state means.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Menu, levels, game over: learners follow the steps.
14 min	**Perform**	Code Lab: start menu, Code Lab: level up
9 min	**Probe**	Game states, activity
4 min	**Wrap-up**	Exit ticket: Which block moves the game from the menu to playing?

Unplugged warm-up

Traffic-light game (5 min): a learner shows red/amber/green cards; the class acts out what each state means.

Teaching the idea

Game states (menu, play, level n, game over) are coordinated with broadcasts (events) and state variables (level, lives). Reset all state on the green flag. This is a lightweight state machine.

Analogy: Traffic lights: the light is always in one state and changes on a signal.

Live demo: Draw a state diagram on the board (menu → play → level 2 → game over) and map each arrow to a broadcast or variable change.

Questions to ask

Ask	Listen for
What starts the play state?	A broadcast (start).
How does the game know the level?	A level variable.
Why reset everything on the flag?	Every game should start fresh.

Differentiation

Support
Provide the menu script; learners add the receive start script.

Stretch
Add a lives variable and a game-over backdrop.

Common misconceptions

Misconception	How to fix it
Every sprite starts on the green flag	In stateful games, many start on receive start.
The level variable makes the game harder by itself	You must use level to change speed or difficulty.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Which block moves the game from the menu to playing?

Homework
Draw your capstone game's states as boxes and arrows.

Answer key — Kitabash activities

1. Code Lab: start menu (perform · code lab)

Goals checked: Menu message · Space broadcasts start · The game begins on start · The Ship flies

One solution:

Ship

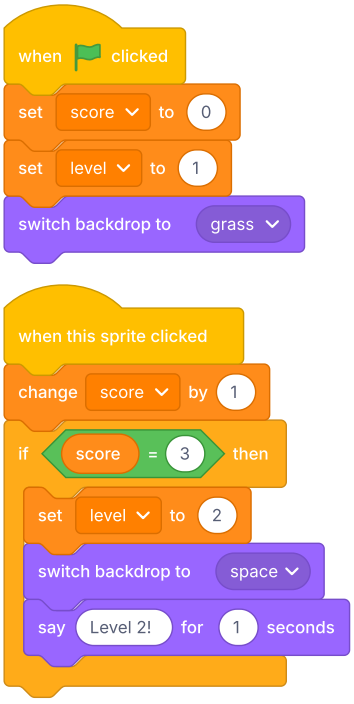


2. Code Lab: level up (perform · code lab)

Goals checked: Reaches level 2 · New backdrop · Fresh start

One solution:

Star



3. Game states (probe · ordering)

Order: Start menu: “Press space to start” → Broadcast start: the game begins → Score reaches the target: level 2 → Time runs out: broadcast game over → Show the final score and stop all

4. Explain your thinking (probe · free response)

Look for: menu, level, game over. **Example:** On the menu my cat says press space. During play, fish fall as clones and I catch them. At 5 points the level changes to night and fish fall faster. At game over the cat says my score and everything stops.

Answer key — worksheet

1. **Key words:** state = What part of the game you are in right now. · level = A stage of a game, usually harder each time. · game over = The state when the game ends.
2. **Plan your code:** Goals: Menu message; Space broadcasts start; The game begins on start; The Ship flies.
3. **Plan your code:** Goals: Reaches level 2; New backdrop; Fresh start.
4. **Game states:** 1 Start menu: “Press space to start” · 2 Broadcast start: the game begins · 3 Score reaches the target: level 2 · 4 Time runs out: broadcast game over · 5 Show the final score and stop all

5. **Explain your thinking:** On the menu my cat says press space. During play, fish fall as clones and I catch them. At 5 points the level changes to night and fish fall faster. At game over the cat says my score and everything stops.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.